

CSC 4710: Database Systems

Spring 2025, Assignment 1

Due date: Friday, 02/13/2026

Problem 1 [20]

Draw the ER diagram for a local bookstore database based on the following information:

1. Each book has a title, abstract, and ISBN. The ISBN is unique to each book;
2. Some of these books are computer-science textbooks, in which case there is an additional piece of associated information — the programming language used for instruction in the book
3. One book may cite other books, and may be cited by other books as well;
4. Each book is published by a publisher although such publisher information may not always be available in the database;
5. A publisher has a unique name, as well as a city and year of the establishment;
6. Each book is written by one or more authors. An author must have written at least one textbook to be included in the database;
7. An author has a name, an address and an SSN. The SSN is unique to each author.

Remember to indicate the key for each entity, as well as the multiplicity of each relationship (e.g. one-to-many) and the referential integrity constraints, using the appropriate notation.

Problem 2 [10]

Based on the diagram you drew in Problem 2, convert this ER diagram into relational model with a set of relations.

Problem 3 [20]

Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars, and has one or more premium payments associated with it. Each payment is for a particular period of time and has an associated due date and the date when the payment was received.

If you feel that you must make some assumptions, please state them clearly. Add required/ necessary attributes to entities. Remember to indicate the key for each entity, as well as the multiplicity of each relationship (e.g. one-to-many) using the appropriate notation.

Problem 4 [10]

Translate the ER diagram of the above problem (problem 3) into a relational model. Specify the primary key and foreign key of each relation in your schema.

Problem 5 [20=5*4]

We have designed an eBay-like on-line auction database. **Answer the following queries with SQL** (Note: Every auction listed in the system, sold, expired, or ongoing, is listed in Auction table. Auctions, whose items have been sold out, are also listed in Sold table.)

- Item (ItemID, ItemName, Description)
- User (Username, Email, Rating)
- Auction (AuctionID, ItemID, SellerUsername, Price, StartTime, EndTime)
- Sold (AuctionID, BuyerUsername, DateSold)

1. Find a distinct list of usernames of the users who participate in an auction. Note that a user may participate in an auction as either a buyer or a seller;
2. Find a list of usernames of the sellers whose items have been sold;
3. Find usernames of the users who have a rating higher than 4.5;
4. Find the ItemName of the highest priced item, along with the Username of the

corresponding user selling the item. Note that there may be more than one item at the highest price, in which case, you must return the information for all items.

